TRANSPORTATION MODELS: VOGEL APPROXIMATION METHOD (VAM)

Forty-fourth Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



Some special items are produced at factories at Patna, Guwahati, Mumbai and Hyderabad. Distributors are located at Lucknow, Delhi, Jaipur, Chennai and Kolkata. Supply and demands of items are indicated in table. Note that factories have capacity to produce 18,000 items, but only 15,000 items are in demand. Transportation cost of each item is proportional to distance it travels. The road distances from factory to distribution centers (in km) are indicated in each cell. Your goal is to minimize the transportation cost.

Distributor Factory	Lucknow	Delhi	Jaipur	Chennai	Kolkata	Supply
Patna	610	1,060	1,180	1,490	580	3,000
Guwahati	1,020	1,910	1,946	2,700	1,018	6,000
Mumbai	1,370	1,420	1,150	1,330	2,050	4,000
Hyderabad	1,300	1,560	1,470	630	1,500	5,000
Demand	2,000	7,000	4,000	1,000	1,000	

Unbalanced Problem: add dummy destination

Add a dummy demand constraint with 0 unit cost and with allocation (18000 – 15000) = 3000.

First allocate cells except dummy column and at last step in dummy column.

Distributor Factory	Lucknow	Delhi	Jaipur	Chennai	Kolkata	Dummy	Supply
Patna	610	1,060	1,180	1,490	580	0	3,000
Guwahati	1,020	1,910	1,946	2,700	1,018	0	6,000
Mumbai	1,370	1,420	1,150	1,330	2,050	0	4,000
Hyderabad	1,300	1,560	1,470	630	1,500	0	5,000
Demand	2,000	7,000	4,000	1,000	1,000	3000	

Stepwise Procedure

Step 1

For each row (column), determine a penalty measure by subtracting the smallest unit cost in the row (column) from the next smallest unit cost in the same row (column).

For example, for row Patna, the smallest unit cost is 0 and the next smallest unit cost is 580. Row penalty is 580 – 0 = 580. Similarly, for column Lucknow, the smallest unit cost is 610 and the next smallest unit cost is 1020. Column penalty is 1020 – 610 = 410.

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply	Row Penalty
Patna	610	1060	1180	1490	580	0	3000	580 = 580 - 0
Guwahati	1020	1910	1946	2700	1018	0	6000	1018 = 1018 - 0
Mumbai	1370	1420	1150	1330	2050	0	4000	1150 = 1150 - 0
Hyderabad	1300	1560	1470	630	1500	0	5000	630 = 630 - 0
Demand	2000	7000	4000	1000	1000	3000		
Column Penalty	410 = 1020 - 610	360 = 1420 - 1060	30 = 1180 - 1150	700 = 1330 - 630	438 = 1018 - 580	0 = 0 - 0		

Step 2

Identify the row or column with the largest penalty. Allocate as much as possible to the cell with the least unit cost in the selected row or column.

The maximum penalty (1150) is in row Mumbai, and the corresponding least unit cost is 0. The maximum allocation in this cell (Mumbai, D_{dummy}) is min (4000, 3000) = **3000**. This satisfies the entire demand of D_{dummy} (cross out the column D_{dummy}) and leaves 4000 – 3000 = 1000 units with Mumbai.

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply	Row Penalty
Patna	610	1060	1180	1490	580	0	3000	30 = 610 - 580
Guwahati	1020	1910	1946	2700	1018	0	6000	2 = 1020 - 1018
Mumbai	1370	1420	1150	1330	2050	0(3000)	1000	180 = 1330 - 1150
Hyderabad	1300	1560	1470	630	1500	0	5000	670 = 1300 - 630
Demand	2000	7000	4000	1000	1000	0		
Column Penalty	410 = 1020 - 610	360 = 1420 - 1060	30 = 1180 - 1150	700 = 1330 - 630	438 = 1018 - 580	--		

Step 3

The next maximum penalty (700) is in column Chennai, and the corresponding least unit cost is 630.

The maximum allocation in this cell (Hyderabad, Chennai) is min (5000,1000) = **1000**. This satisfies the entire demand of Chennai (cross out the column Chennai) and leaves 5000 – 1000 = 4000 units with Hyderabad.

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply	Row Penalty
Patna	610	1060	1180	1490	580	0	3000	30 = 610 - 580
Guwahati	1020	1910	1946	2700	1018	0	6000	2 = 1020 - 1018
Mumbai	1370	1420	1150	1330	2050	0(3000)	1000	220 = 1370 - 1150
Hyderabad	1300	1560	1470	630(1000)	1500	0	4000	170 = 1470 - 1300
Demand	2000	7000	4000	0	1000	0		
Column Penalty	410 = 1020 - 610	360 = 1420 - 1060	30 = 1180 - 1150	--	438 = 1018 - 580	--		

Step 4

The next maximum penalty (438) is in column Kolkata, and the corresponding least unit cost is 580.The maximum allocation in this cell (Patna, Kolkata) is min (3000,1000) = **1000**. This satisfies the entire demand of Kolkata (cross out the column Kolkata) and leaves 3000 – 1000 = 2000 units with Patna.

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply	Row Penalty
Patna	610	1060	1180	1490	580(1000)	0	2000	450 = 1060 - 610
Guwahati	1020	1910	1946	2700	1018	0	6000	890 = 1910 - 1020
Mumbai	1370	1420	1150	1330	2050	0(3000)	1000	220 = 1370 - 1150
Hyderabad	1300	1560	1470	630(1000)	1500	0	4000	170 = 1470 - 1300
Demand	2000	7000	4000	0	0	0		
Column Penalty	410 = 1020 - 610	360 = 1420 - 1060	30 = 1180 - 1150	--	--	--		

Step 5

The next maximum penalty (890) is in row Guwahati, and the corresponding least unit cost is 1020. The maximum allocation in this cell (Guwahati, Lucknow) is min (6000, 2000) = 2000. This satisfies the entire demand of Lucknow (cross out the column Lucknow) and leaves 6000 – 2000 = 4000 units with Guwahati.

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply	Row Penalty
Patna	610	1060	1180	1490	580(1000)	0	2000	120 = 1180 - 1060
Guwahati	1020(2000)	1910	1946	2700	1018	0	4000	36 = 1946 - 1910
Mumbai	1370	1420	1150	1330	2050	0(3000)	1000	270 = 1420 - 1150
Hyderabad	1300	1560	1470	630(1000)	1500	0	4000	90 = 1560 - 1470
Demand	0	7000	4000	0	0	0		
Column Penalty	--	360 = 1420 - 1060	30 = 1180 - 1150	--	--	--		

Step 6

The next maximum penalty (360) is in column Delhi, and the corresponding least unit cost is 1060. The maximum allocation in this cell (Patna, Delhi) is min (2000, 7000) = 2000. This exhausts the entire supply of Patna (leaves 2000 – 2000 = 0 unit with Patna, cross out the row Patna) and leaves 7000 – 2000 = 5000 units with Delhi.

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply	Row Penalty
~~Patna~~	~~610~~	~~1060(2000)~~	~~1180~~	~~1490~~	~~580(1000)~~	~~0~~	0	--
Guwahati	1020(2000)	1910	1946	2700	1018	0	4000	36 = 1946 - 1910
Mumbai	1370	1420	1150	1330	2050	0(3000)	1000	270 = 1420 - 1150
Hyderabad	1300	1560	1470	630(1000)	1500	0	4000	90 = 1560 - 1470
Demand	0	5000	4000	0	0	0		
Column Penalty	--	140 = 1560 - 1420	320 = 1470 - 1150	--	--	--		

Step 7

The next maximum penalty (320) is in column Jaipur, and the corresponding least unit cost is 1150. The maximum allocation in this cell (Mumbai, Jaipur) is min (1000, 4000) = 1000. This exhausts the entire supply of Mumbai (leaves 1000 – 1000 = 0 unit with Mumbai, cross out the row Mumbai) and leaves 4000 – 1000 = 3000 units with Jaipur.

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply	Row Penalty
~~Patna~~	~~610~~	~~1060(2000)~~	~~1180~~	~~1490~~	~~580(1000)~~	~~0~~	0	--
Guwahati	1020(2000)	1910	1946	2700	1018	0	4000	36 = 1946 - 1910
~~Mumbai~~	~~1370~~	~~1420~~	~~1150(1000)~~	~~1330~~	~~2050~~	~~0(3000)~~	0	--
Hyderabad	1300	1560	1470	630(1000)	1500	0	4000	90 = 1560 - 1470
Demand	0	5000	3000	0	0	0		
Column Penalty	--	350 = 1910 - 1560	476 = 1946 - 1470	--	--	--		

Step 8

The maximum penalty (476) is in column Jaipur, and the corresponding least unit cost is 1470. The maximum allocation in this cell (Hyderabad, Jaipur) is min (4000, 3000) = 3000. This satisfies the entire demand of Jaipur (cross out the column Jaipur) and leaves 4000 – 3000 = 1000 units with Hyderabad.

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply	Row Penalty
~~Patna~~	~~610~~	~~1060(2000)~~	~~1180~~	~~1490~~	~~580(1000)~~	~~0~~	0	--
Guwahati	1020(2000)	1910	1946	2700	1018	0	4000	1910
~~Mumbai~~	~~1370~~	~~1420~~	~~1150(1000)~~	~~1330~~	~~2050~~	~~0(3000)~~	0	--
Hyderabad	1300	1560	1470(3000)	630(1000)	1500	0	1000	1560
Demand	0	5000	0	0	0	0		
Column Penalty	--	350 = 1910 - 1560	--	--	--	--		

Step 9

The next maximum penalty (1910) is in row Guwahati, and the corresponding least unit cost is 1910. The maximum allocation in this cell (Guwahati, Delhi) is min (4000, 5000) = 4000. This exhausts the entire supply of Guwahati (cross out the row Guwahati) and leaves 5000 – 4000 = 1000 units with Delhi.

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply	Row Penalty
~~Patna~~	~~610~~	~~1060(2000)~~	~~1180~~	~~1490~~	~~580(1000)~~	~~0~~	0	--
~~Guwahati~~	~~1020(2000)~~	~~1910(4000)~~	~~1946~~	~~2700~~	~~1018~~	~~0~~	0	--
~~Mumbai~~	~~1370~~	~~1420~~	~~1150(1000)~~	~~1330~~	~~2050~~	~~0(3000)~~	0	--
Hyderabad	1300	1560	1470(3000)	630(1000)	1500	0	1000	1560
Demand	0	1000	0	0	0	0		
Column Penalty	--	1560	--	--	--	--		

Step 10

We are left with the cell (Hyderabad, Delhi). The maximum allocation in this cell is min (1000, 1000) = 1000. This exhausts the entire supply of Hyderabad (cross out the row Hyderabad) and satisfies the entire demand of Delhi (cross out the column Delhi).

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}
Patna	610	1060(2000)	1180	1490	580(1000)	0
Guwahati	1020(2000)	1910(4000)	1946	2700	1018	0
Mumbai	1370	1420	1150(1000)	1330	2050	0(3000)
Hyderabad	1300	1560	1470(3000)	630(1000)	1500	0

Step 11

We have reached the final initial allocation. The initial feasible solution is

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610	1060(**2000**)	1180	1490	580(**1000**)	0	3000
Guwahati	1020(**2000**)	1910(**4000**)	1946	2700	1018	0	6000
Mumbai	1370	1420	1150(**1000**)	1330	2050	0(**3000**)	4000
Hyderabad	1300	1560(**1000**)	1470(**3000**)	630(**1000**)	1500	0	5000
Demand	2000	7000	4000	1000	1000	3000	

Total minimum transportation cost

=1060×2000 + 580×1000 +1020×2000 + 1910×4000 + 1150×1000 + 0×3000 +1560×1000 +1470×3000 + 630×1000 = 20,130,000.

Vogel approximation method (VAM):

VAM is an improved version of the least-cost method that generally, but not always, produces better starting solutions.

- **Step 1:** For each row (column), determine a penalty measure by subtracting the smallest unit cost in the row (column) from the next smallest unit cost in the same row (column). This penalty is actually a measure of lost opportunity one for-goes if the smallest unit cost cell is not chosen.
- **Step 2:** Identify the row or column with the largest penalty, breaking ties arbitrarily. Allocate as much as possible to the variable with the least unit cost in the selected row or column. Adjust the supply and demand, and cross out the satisfied row or column. If a row and a column are satisfied simultaneously, only one of the two is crossed out, and the remaining row (column) is assigned zero supply (demand).

Vogel approximation method (VAM):

- **Step 3:** (a) If exactly one row or column with zero supply or demand remains un-crossed out, stop.

  (b) If one row (column) with positive supply (demand) remains uncrossed out, determine the basic variables in the row (column) by the least-cost method. Stop.

  (c) If all the uncrossed-out rows and columns have (remaining) zero supply and demand, determine the zero basic variables by the least-cost method. Stop.

  (d) Otherwise, go to step 1.

  (Based on book by H.A. Taha).

Destination Source	D1	D2	D3	D4	D5	Supply
S1	3	4	6	8	9	30
S2	2	2	4	5	5	80
S3	2	2	2	3	2	10
S4	3	3	2	4	2	20
Demand	10	50	20	40	20	

Destination Source	D1	D2	D3	D4	D5	Supply	Penalty
S1	3	4	6	8	9	30	1
S2	2	2	4	5	5	80	0
S3	2	2	2	3	2	10	0
S4	3	3	2	4	2	20	0
Demand	10	50	20	40	20		
Penalty	0	0	0	1	0		

Destination Source	D1	D2	D3	D4	D5	Supply	Penalty
S1	10	4	6	8	9	30	2
S2	0	2	4	5	5	80	2
S3	0	2	2	3	2	10	0
S4	0	3	2	4	2	20	0
Demand	10	50	20	40	20		
Penalty		0	0	1	0		

Destination Source	D1	D2	D3	D4	D5	Supply	Penalty
S1	10	20	0	0	0	30	
S2	0	2	4	5	5	80	2
S3	0	2	2	3	2	10	0
S4	0	3	2	4	2	20	0
Demand	10	50	20	40	20		
Penalty		0	0	1	0		

Destination Source	D1	D2	D3	D4	D5	Supply	Penalty
S1	10	20	0	0	0	30	
S2	0	30	4	5	5	80	1
S3	0	0	2	3	2	10	0
S4	0	0	2	4	2	20	0
Demand	10	50	20	40	20		
Penalty			0	1	0		

Destination Source	D1	D2	D3	D4	D5	Supply	Penalty
S1	10	20	0	0	0	30	
S2	0	30	20	5	5	80	0
S3	0	0	0	3	2	10	1
S4	0	0	0	4	2	20	2
Demand	10	50	20	40	20		
Penalty				1	0		

Destination Source	D1	D2	D3	D4	D5	Supply	Penalty
S1	10	20	0	0	0	30	
S2	0	30	20	5	0	80	
S3	0	0	0	3	0	10	
S4	0	0	0	0	20	20	
Demand	10	50	20	40	20		
Penalty							

Destination Source	D1	D2	D3	D4	D5	Supply	Penalty
S1	10	20	0	0	0	30	
S2	0	30	20	30	0	80	
S3	0	0	0	10	0	10	
S4	0	0	0	0	20	20	
Demand	10	50	20	40	20		
Penalty							

Transportation cost $=3\times10+4\times20+2\times30+4\times20+5\times30+3\times10+2\times20=470$